The I_{3322} quantum value is attained spatially but not in finite dimension

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Abstract

Let $S = \omega_{\text{tensor}}(I_{3322}) = \omega_{\text{commuting}}(I_{3322})$ denote the quantum supremum of the I_{3322} Bell functional in the Collins–Gisin normalization (classical bound 0; two-qubit maximum exactly 1/4). Working from an exactly certified window $S \in (0.2508753845015185, 0.250875388108398]$ and a certified equality of the tensor-product and commuting-operator suprema, we prove: (i) **no finite-dimensional quantum strategy attains** S — any finite local dimensions, pure or mixed states, projective or POVM measurements — establishing the finite-dimensional half of the conjecture of Pál and Vértesi (2010); (ii) S **is** attained by a normal spatial strategy on $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$. Consequently $C_q(3,3;2,2)$ is not closed — the smallest scenario by input count, among two-outcome bipartite scenarios, in which nonclosure is known — and $C_{qs}(3,3;2,2) \setminus C_q(3,3;2,2) \neq \emptyset$, discharging the premise of an implication recorded by Dykema, Paulsen, and Prakash. The nonattainment proof runs through a concave critical Bellman storage, exact rational endpoint-exclusion certificates, a reflection-gluing inequality, and a convex-envelope theorem excluding contact plateaus; finiteness then forces the two equality transports of an exact maximizer to coincide, which caps its value at $1/4 < S$. The attainment proof disintegrates the scalar spectral measure of a maximizing commuting state over the orbits of its response translation and reads off an ℓ^2 Jacobi eigenvector, with normalizability inherited from the probability measure rather than imposed. The load-bearing steps were audited in preregistered refutation-first adversarial review conducted by independently instantiated frontier language models (see the disclosure in Section 6), whose signed final records ship with the certificates, together with a full correction history: an earlier claimed proof by this program was publicly decertified after an exact audit, and nothing decertified re-enters the present chain. We close with a structural observation suggested by the results: in every case known to us, finite-dimensional attainment coincides with closed (finitely recurrent) optimal carriers, while the known transport-class nonattainment phenomena are carried by infinite open chains.

1 Introduction

The I_{3322} inequality [1, 2] is the best-known bipartite Bell functional beyond the Clauser–Horne–Shimony–Holt family: three binary measurements per party, and a facet of the local polytope inequivalent to CHSH. In 2010, Pál and Vértesi [3] computed its quantum value numerically to high precision, exhibited a family of finite-dimensional strategies converging to $0.2508753845\ldots$, observed that the qubit maximum is exactly $1/4$ and that within their construction no local dimension below 12 exceeds it, and posed the following, verbatim:

> "Taking as a conjecture that our construction is optimal (in the sense that no finite dimension suffices to achieve the true quantum maximum for I_{3322}), to our knowledge this constitutes the first example of a [bipartite] Bell scenario [with a] finite number of measurements and outcomes, where measuring finite dimensional quantum systems is not enough to obtain exactly all quantum correlations. ... We pose it as a challenge to provide an analytical proof of our conjecture." [3]

The conjecture has two parts: that no finite dimension attains the supremum, and that their particular construction is optimal. This paper proves the first part, in both directions of the attainment question; the optimality of the Pál–Vértesi construction, and the exact identification of the value, remain open (Section 5). Coladangelo and Stark, in constructing the first explicit bipartite correlation attainable only in infinite dimension (on larger alphabets), wrote of the I_{3322} case that "an analytical proof has remained elusive" [4] (we quote the Nature Communications version).

Theorem 1 (Nonattainment)**.** *Let $S = \omega_{\text{tensor}}(I_{3322}) = \omega_{\text{commuting}}(I_{3322})$ be the quantum supremum in the normalization of Section 2, certified to lie in* $(0.2508753845015185,\ 0.250875388108398]$*. Then no finite-dimensional quantum strategy attains S: for all finite-dimensional Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$, every (pure or mixed) state on $\mathcal{H}_A \otimes \mathcal{H}_B$ and every triple of binary measurements per party (projective or POVM), $\langle I_{3322}\rangle < S$.*

Theorem 2 (Spatial attainment)**.** *S is attained by a normal spatial strategy: there is a unit vector $\psi_S \in \ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ and six projective binary measurements, built in the alternating block* form *of the Pál–Vértesi construction but with block labels and amplitudes derived from a maximizing commuting state (Section 4), with $\langle \psi_S, \mathcal{B}_{3322}\, \psi_S\rangle = S$.*

Here C_q denotes the set of bipartite correlations realizable with finite-dimensional tensor-product strategies, C_{qs} its infinite-dimensional tensor-product analogue, and $C_{qa} = \overline{C_q}$; see [19, 4] for the standard taxonomy.

Corollary 3. *$C_q(3,3;2,2)$ is not closed, and $C_{qs}(3,3;2,2) \setminus C_q(3,3;2,2) \neq \emptyset$.*

Dykema, Paulsen, and Prakash observed in 2018 that conjectured I_{3322} nonattainment would imply nonclosure of $C_q(3,2)$ [7]; the corollary discharges the premise of that previously recognized implication. Among *two-outcome bipartite* scenarios, $(3,3;2,2)$ is the smallest venue by input count in which nonclosure is known: the previously smallest examples are the $(5,5;2,2)$ synchronous construction of Dykema–Paulsen–Prakash [6] (with a simplified proof by Musat–Rørdam [8]) and Beigi's $(4,4;2,2)$ separation of C_q from C_{qs} together with his $(4,4;3,3)$ nonclosure [9]; Coladangelo–Stark's correlation lives at $(4,5;3,3)$ [4], and the linear-system game of [11] has input sets of sizes 184 and 235 with output sets of sizes 8 and 2. In the $(2,2;2,2)$ scenario every functional attains its maximum on two qubits [12], so three inputs is also the smallest possible venue among two-outcome scenarios. Coladangelo and Stark had already singled this scenario out: "the $(3,3,2,2)$ scenario is the simplest one that is suspected to separate C_q and C_{qs}" [4]. We do not claim minimality over scenarios with larger output alphabets.

Correction history, stated first

This program previously released a claimed proof of the I_{3322} headline (exact optimum, spatial attainment, nonattainment) whose load-bearing global amplitude datum was subsequently *refuted by our own post-release exact audit*: the amplitude-compatibility equation of the original bi-infinite wall construction fails by an exactly certified margin (the mismatch lies in $[1.40 \times 10^{-4}, 1.79 \times 10^{-4}]$, excluding zero, reproduced by two independent interval engines). The claims were publicly decertified within a day, and the repository's release history preserves the full correction record. The results announced here were then rebuilt on independent routes; the decertified fixed point, amplitude profile, and tail constants appear on explicit *not-used* lists in the certificate dependency records, and the failed compatibility equation is not repaired anywhere in the present chain — the new attainment route never poses it (Section 4). The load-bearing theorems below then survived three rounds (Theorem 1) plus one round (Theorem 2) of preregistered, refutation-first adversarial review; the nature of that review, its limitations, and the disposition of reviewer-supplied repairs are disclosed in Section 6. We state this history first because computer-assisted claims are only as credible as their failure handling; the correction record is part of the result.

2 Setting and certified inputs

For projections $A_1, A_2, A_3, B_1, B_2, B_3$, the Bell operator in the convention used throughout the repository is

$$\mathcal{B} \;=\; -A_2 - B_1 - 2B_2 \;+\; A_1B_1 + A_1B_2 - A_1B_3 \;+\; A_2B_1 + A_2B_2 + A_2B_3 \;-\; A_3B_1 + A_3B_2, \quad (1)$$

which coincides, with no scaling or additive shift, with Eqs. (9)–(14) of Pál–Vértesi [3]; a permutation of settings brings it to the standard Collins–Gisin table with local (classical deterministic) bound 0 [2]. The two-qubit maximum is exactly $1/4$ [3]; Pál–Vértesi further report, as a numerical finding of their dimension scan, that no construction they found below local dimension 12 exceeds $1/4$, with their strategy family converging to $0.250875384514\ldots$ [3]. Throughout, S denotes the common quantum value of Theorem 1, identified *only* by its certified window; we do not claim S equals the historical decimal, nor that the Pál–Vértesi family is optimal, though all certified data are consistent with both.

Exact rational two-sided enclosures for the I_{3322} quantum value predate this program's releases: Mghirbi [5] published proof-carrying exact certificates with enclosure width below 10^{-9} — tighter than the window used here — in July 2026. Those certificates establish a strict enclosure; they do not address equality of the two models, attainment, or nonattainment, which are the subject of this paper. The window below is therefore a certified *input*, not a contribution.

The proofs consume the following exactly certified inputs, established in the public repository [29]: (T0) a Bellman/path variational theorem giving $\omega_{\text{tensor}} = \omega_{\text{commuting}} = S$, together with, for every level $q > S$, a terminal-pivot storage g_q — the infimum of terminal Schur pivots over finite histories — which is positive, Bellman-feasible, and equicontinuous in q; (LB) an exact 255-dimensional strategy certifying $S > 0.2508753845015185 > 1/4$ (rational square-root floors, independently reconstructed in 160-digit interval arithmetic); (UB) an exact rational 25,601-knot Bellman witness certifying $S \leq 0.250875388108398$; and (W) a generic operator weld (sprint 1287, building on the local remainder structure of sprint 1197): any positive Bellman-feasible storage at level q yields a positive decomposition $qI - \mathcal{B} = R_0 + R_A + R_B$. The finite Pál–Vértesi block-to-Jacobi identity (sprint 1206, §§2–4 only — the finite, endpoint-free form) supplies the bridge between Jacobi data and Bell strategies; the adversarial review independently replicated it at machine precision.

3 Nonattainment

We sketch the architecture; complete proofs, with every constant exact, are in the certificate documents [29], directory `theorem-N-four-receipts-at-S`.

The critical storage. As $q \downarrow S$ the storages g_q admit, by equicontinuity and Arzelà–Ascoli, a subsequence converging uniformly to a limit g, which is continuous, nonnegative, *concave* (each terminal pivot is affine in the target label, and an infimum of affine functions is concave), positive on the open interval, and Bellman-feasible at S. Only these properties are consumed below. Concavity is elementary but decisive: it forces every corner of the contact intercept downward, which is what ultimately excludes contact plateaus.

Exact endpoint exclusion. Two-edge finite histories give exact rational margins: with $r = 1/10$, the endpoint predecessor lines exceed the storage uniformly by

$$m_+ = \tfrac{23686917837403}{3008753881083980} > 0.00787, \qquad m_- = \tfrac{274562305945801}{4008753881083980} > 0.0684,$$

for every level in the certified window; only the positivity $m_\pm > 0$ is used below. Consequently the remainder R_0 has a uniform positive gap on all endpoint spectral sectors, so no exact or near-maximizing state carries endpoint mass, and $x = \pm 1$ is never a Bellman predecessor at S. These are closed-form rational certificates; no branch geometry or reflected wing is used.

The equality module of a hypothetical maximizer. Suppose a finite-dimensional strategy attains S. (Purification is unnecessary: $\mathrm{Tr}(\rho R) = 0$ with $R \succeq 0$ gives $R\rho = 0$ directly. POVMs reduce to projective measurements by the extreme-effect replacement of the certificate documents: holding the state and the other effects fixed, each binary effect may be replaced by an extreme maximizer of its linear objective, and extreme binary effects are projections; this preserves the dimension, the tensor structure, and the value.) Applying the weld at levels $q_n \downarrow S$ and using $\sum_\nu \mathrm{Tr}(\rho R_{\nu,n}) = q_n - S \to 0$, the endpoint gaps kill boundary mass first; on the remaining interior finite spectrum the coefficients converge, and the limiting positive remainders annihilate ρ. Thus the maximizer carries an exact critical equality module at S — *without* any claim that the storage infimum is attained (the endpoint boundary layer is bypassed, not solved).

Zero-set geometry. On the equality module, the scalar remainder symbol vanishes. Reflected finite histories give the gluing inequality $g(x)\, g(-x) \geq b(x)^2$ with $b(x)^2 = (1-x^2)/4$; full equality of the two Bellman inequalities and the Cauchy–Schwarz step forces $g(x)g(-x) = b(x)^2$ at every occupied pair, and a Monge (supermodularity) argument makes the full zero set monotone. The decisive step is a convex-envelope theorem: writing $C(x) = S + 1 - \frac{x}{2} - b(x)^2/g(x)$ on the open interval $(-1, 1)$ — where $C \geq g(0) > 0$ by feasibility at the origin, so the greatest convex minorant H exists — concavity of g makes every corner of C point downward, while a kink of H at a contact point would force an upward corner. Hence H is differentiable, no source serves two targets, and (by an exact involution $(x, u) \mapsto (-u, -x)$ of the constraint system, which requires no symmetry of g itself) no target is served by two sources. The occupied support of any maximizer is therefore a strictly increasing one-to-one graph.

Finiteness forces closure; closure caps the value. The equality module carries two unitary response transports implementing the decreasing involutions $a(u) = P^{-1}(-P(u))$ and $b(u) = -u$ on the occupied support; the graph property makes both total. A finite totally ordered set admits exactly one decreasing bijection, so $a = b$ on the support: every occupied pair has its reflected partner. The resulting amplitude-ratio elimination (multiplicity-uniform: the transports are scalar multiples of unitaries between complete fibres) yields, for an occupied pair (x, u),

$$S \;=\; xu - 1 + \sqrt{(b_x + b_u)^2 + \tfrac{(x-u)^2}{4}} \;\leq\; -t + \sqrt{t} \;\leq\; \tfrac{1}{4}, \qquad t = 1 - xu,$$

contradicting $S > 1/4$. This proves Theorem 1. The nonclosure half of Corollary 3 follows by compactness of the behavior space: finite-dimensional behaviors approach value S; a convergent subsequence has value exactly S; closedness of C_q would give a finite-dimensional realization attaining S.

Remark 4. The only role of finite dimension is the single sentence "a finite ordered set has one decreasing bijection." Everything else holds for arbitrary maximizers, which is exactly why Theorem 2 is possible.

4 Spatial attainment

The historical route to attainment constructed amplitudes along a bi-infinite wall and then demanded a global compatibility equation — the equation whose exact failure decertified the original release. The present route *dissolves* that obstruction rather than repairing it: no amplitude is ever constructed, and the equation is never posed.

By weak-$*$ compactness of the state space of the universal commuting C^*-algebra, a maximizing state Ω exists for the commuting model: the commuting-operator value of a Bell functional is always a maximum [17, 18], so nonattainment is strictly a tensor-model phenomenon. The limit passage of Section 3 then applies with one essential modification: no finite spectral support is available, and the transports are expressed through the exact operator blocks $W = Y(B_3 - \frac{1}{2}I)$ and $W_B = (A_3 - \frac{1}{2}I)V$, which satisfy $W^2 = b(X)^2$ and $WX = -XW$ as projector identities (no global spectral involution is assumed; it need not exist). These yield exact Radon–Nikodym transport laws for the scalar spectral measure μ of the commuting pair (X, U). The fixed-point set of the response translation $\tau = a \circ b$ carries no mass (on it the two transport densities coincide, which fires the quarter-ceiling elimination against $S > 1/4$), so μ disintegrates over the countable free orbits of τ via an explicit Borel transversal. Choosing one conditional orbit measure and interleaving the weights of $\{u_n\}$ and $\{-u_n\}$ produces amplitudes $\lambda_j = \sqrt{\mu_t(\{c_j\})}$ with, automatically, $1 \leq \sum_j \lambda_j^2 \leq 2$: *normalizability is inherited from the probability measure, not imposed* — this is the precise point where the historical obstruction disappears. The transport laws then *derive* the cocycle $\lambda_{j+1}/\lambda_j = g(c_j)/b(c_j)$, and the Bellman contact equality closes the Jacobi eigenvalue equation $H\lambda = S\lambda$ exactly. The finite Pál–Vértesi block identity installs the normalized eigenpair as a spatial strategy with value S — the block *form* is Pál–Vértesi's; the labels c_j and amplitudes λ_j come from the disintegration — proving Theorem 2 and, with Theorem 1, the separation in Corollary 3.

Remark 5 (Envelope versus orbit). The closure argument of Section 3 forces the $u \mapsto -u$ symmetry only on the *occupied orbit* of a hypothetical finite maximizer; nothing forces the critical envelope itself to be reflection-symmetric, and no step above obtains one wing of the envelope by reflecting the other. The historical construction's implicit reflection symmetry is, in retrospect, adjacent to the exact incompatibility that decertified it.

5 Discussion: attainment and open carriers

The two theorems exhibit a sharp mechanism: *finiteness forces the equality transports to close (return), and closure caps the value strictly below S; the value above the closed ceiling is carried only by an infinite open chain.* It is natural to ask how general this picture is. In every case known to us, a Bell functional with proven finite-dimensional attainment of its global supremum closes for an identifiable algebraic reason: bipartite correlator (XOR) functionals — including the chained inequalities — via Tsirelson's Clifford construction [13, 14] (anticommutation is two-step closure); all $(2, 2; 2, 2)$ functionals, whose extreme points are realized by qubit strategies [12]; fixed CGLMP functionals, whose global optimum is attained at local dimension d (proven for $d = 3, 4$ [15]; we make no minimality claim); and the families certified by sum-of-squares decompositions that force finite-dimensional algebraic relations on optimal strategies. Conversely, both known explicit transport-class nonattainment objects are infinite open ladders: the Coladangelo–Stark state is an exactly geometric ladder whose nonattainment proof (Theorem 12 of arXiv:1804.05116) is an explicit no-return orbit argument on the Schmidt spectrum [4], and the Pál–Vértesi family is a half-infinite Jacobi chain with offset 2×2 blockings [3] — the same architecture, on smaller alphabets. Coladangelo and Stark suggested that the block structure of their construction makes I_{3322} "potentially amenable to ideas and techniques from our work"; Theorem 1 may be read as an answer to that suggestion, reached by a transfer-operator route. Embezzlement-based nonattainment [20, 21, 22] is carried by the log-flat ladder $p_j \propto 1/j$ — an open chain of a different decay class; whether this heuristic correspondence with its operator-algebraic classification [23] can be made precise is an open question we do not pursue here. Group-theoretic and type-based obstructions [10, 6, 8, 24] form a third family outside this transport picture; Slofstra's undecidability of perfect-strategy existence for linear-system games [10], the RE-completeness of approximating the quantum value [24], and the arithmetical-hierarchy separation of Mousavi–Nezhadi–Yuen [25] — exact quantum value Π_2-complete versus commuting value Π_1, precisely because the finite-dimensional supremum need not be attained — imply that any general structural criterion must be restricted to a subclass. Within the two-outcome bipartite class, where Jordan's lemma supplies the 2×2 chain structure, we record the dichotomy as a conjecture: *finite-dimensional attainment coincides with finite recurrence of the critical transport.* A concrete test: the Dykema–Paulsen–Prakash $(5, 5; 2, 2)$ nonclosure [6] is proved non-constructively; the conjecture predicts a forced-open orbit in its critical transport. We also note the marginal/correlator boundary: Tsirelson's theorem covers correlators only, and it is exactly the marginal terms through which I_{3322} escapes it. Finally, two cautions from the literature, both concerning the *hierarchy* sense of "attain" rather than the strategy sense used in this paper: non-saturation of finite NPA levels is not evidence of strategy-nonattainment (the doubly-tilted CHSH functionals are attained at $d = 2$ yet NPA level 10 does not saturate them [26]), and the NPA hierarchy need not attain the commuting value at any finite level [27] (whose headline result is the undecidability of deciding $\omega_{qc} > 1/2$).

What remains open here: the exact identification of S beyond the certified window; the optimality of the Pál–Vértesi construction; and the matching dimension-necessity lower bound (how fast $S - S_d \to 0$ from below); a constructive upper profile with logarithmic dimension sufficiency is part of ongoing work.

6 Methods, verification, and disclosure

Exact computation

All load-bearing computations are exact (rational arithmetic or outward-rounded interval arithmetic) and two-engine where numerical (independent reconstructions in separate codebases; e.g. the 255-dimensional lower strategy is verified in exact rationals and reproduced at 160 digits). They ship as machine-checkable certificates in the public repository [29], with the complete correction record. Package guard scripts check algebraic identities and exact arithmetic only;

the analytic steps are established in the reviewed proof documents, not by the scripts, and the abstract's phrase "computer-assisted" should be read accordingly.

AI disclosure and the nature of the adversarial review

Frontier large language models were used extensively throughout this work: for proof discovery, implementation, verification engineering, adversarial auditing, and editorial assistance. The models included OpenAI GPT-class models (including Codex-class coding models and GPT-5.x), and Anthropic Claude models (Opus-class and successor frontier models), orchestrated across separate, independently instantiated sessions. The "adversarial review" referred to throughout is a *preregistered refutation-first audit protocol executed by independently instantiated frontier language models with no access to the authors' working notes*, each instructed to break the proofs and rewarded (procedurally) for genuine refutations; the first round did refute an earlier restoration attempt with an explicit countermodel, and the final rounds signed the theorems subject to binding claim boundaries that this paper follows. This provides *procedural independence, not ontological independence*: the reviewing models share a broad frontier-model lineage with the models that produced the proofs and may share training priors. It is adopted because it is reproducible, auditable (the complete signed verdicts of the final rounds ship in the certificate directories [29]), and stronger than no review; it is not equivalent to review by human experts, and no such review has yet taken place. Three repair steps in the audited chain were *supplied by the reviewing model* rather than the authors' track and, at this writing, have been verified within the same protocol but not by a fresh reviewer; they are identified in the certificate review records. The human author conceived the research program and its central hypotheses, provided the interpretive direction and redirections, set the verification standards and the preregistration and correction protocols, made all promotion, retraction, and publication decisions, and assumes sole responsibility for this work.

Data availability

Repository: `https://github.com/Apsiape/i3322-exact-wall` (release v3.2.2). Concept DOI (resolves to the current archived release, which includes the certificate directories and review records): 10.5281/zenodo.21782008. Certificate directories: `certificate/production/theorem-N-four-receip` and `certificate/production/theorem-S-spatial-attainment-at-S/`.

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